NON-SMOOTHABLE CALABI–YAU THREEFOLDS FROM REFLEXIVE POLYTOPES

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ABSTRACT. We give explicit examples of compact Calabi–Yau threefolds with isolated Gorenstein canonical singularities that admit no global smoothing. The examples are generic anticanonical hypersurfaces in Gorenstein Fano toric fourfolds; their singular points are cones over two-dimensional faces of the corresponding reflexive 4-polytopes, and smoothability of such cones is governed by Altmann's description of the versal deformation together with the smoothing criterion of Corti–Filip–Petracci: the cone over a lattice polygon with primitive (*unit*) edges is smoothable if and only if the polygon decomposes as a Minkowski sum of unit segments and standard triangles. Consequently, if a reflexive 4-polytope has a two-dimensional face with primitive edges whose cone is not smoothable, then the generic anticanonical hypersurface admits no smoothing: a combinatorial test on the polytope alone, needing no local-to-global obstruction theory. Our first example has exactly three singular points, each isomorphic to the anticanonical cone over the Hirzebruch surface $\mathbb{F}_1$, the exceptional local model in Gross's classification of primitive type II contractions. Our second example has exactly three singular points whose miniversal deformation spaces are positive-dimensional, so every singular point deforms nontrivially, and yet the threefold admits no smoothing; to our knowledge this phenomenon has not previously been exhibited on a compact Calabi–Yau threefold. A variant of the construction produces a Calabi–Yau threefold whose singular locus is a *single* point, the anticanonical cone over $\mathbb{F}_1$; and a scan of the Kreuzer–Skarke classification produces one whose singular locus is a single point of the deformable type. Carried over the whole classification, the scan shows that 8.27% of all 473,800,776 reflexive 4-polytopes carry a unit-edge non-smoothable two-dimensional face, so that non-smoothable Calabi–Yau hypersurfaces occur at scale throughout the Batyrev landscape. Along the way we classify the 217 isolated Gorenstein toric threefold singularities whose polygon has primitive edge vectors that can all be chosen with coordinates in $\{-2, \dots, 2\}$: each is rigid, deformable but non-smoothable, or smoothable, and we record the number of smoothing components in each case.

1. INTRODUCTION

Let X be a projective threefold with trivial dualizing sheaf, $h^1(\mathcal{O}_X) = 0$, and Gorenstein canonical singularities; we call such an X a *Calabi–Yau threefold* with canonical singularities. A basic question, going back to the foundational work of Friedman [8], is whether X can be *smoothed*: does there exist a flat proper family over a pointed disc with special fibre X and smooth general fibre?

For the mildest singularities the question is completely understood. If X has ordinary double points, Friedman [8] showed that a smoothing exists if and only if the exceptional curves of a small resolution satisfy a non-trivial "good" relation in homology; if X has arbitrary isolated terminal Gorenstein singularities, Namikawa and Steenbrink [18] proved that X is always smoothable. Beyond the terminal case the picture changes qualitatively. Gross [13]

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analyzed the case where X arises from a primitive type II contraction of a smooth Calabi–Yau threefold (a divisor E contracted to a point) and proved that X is then smoothable *unless* $E \cong \mathbb{P}^2$ or $E \cong \mathbb{F}_1$ [13, Theorem 5.8]. The two exceptions are genuinely non-smoothable: the germ at the singular point is the anticanonical cone over E, which for $E = \mathbb{P}^2$ is the rigid quotient singularity $\frac{1}{3}(1,1,1)$ [20], and for $E = \mathbb{F}_1$ has one-dimensional tangent space T^1 but no genuine deformations at all [13, Example 2.19]. Deformations of Calabi–Yau threefolds with canonical and log canonical singularities remain an active subject; see Friedman–Laza [9, 10] for recent progress.

This note produces explicit *compact* examples beyond the quotient case, by a mechanism that is purely combinatorial. The inputs are:

(1) Altmann's theorem [2] that for an isolated Gorenstein toric threefold singularity (the cone $C(Q)$ over a lattice polygon Q with primitive edges) the irreducible components of the reduced miniversal base correspond to the maximal decompositions of Q into Minkowski sums of lattice polytopes;
(2) the smoothing criterion of Corti–Filip–Petracci [4, §5.1]: the *smoothing* components correspond to the Minkowski decompositions of Q into *unit segments* and *standard triangles*; consequently $C(Q)$ is smoothable if and only if such a decomposition exists;
(3) Batyrev's construction [3]: for a reflexive polytope $\Delta \subset \mathbb{Z}^4$ (throughout, the *fan* polytope; see §3), the generic anticanonical hypersurface X in the Gorenstein Fano toric fourfold $\mathbb{P}_\Delta$ is a Calabi–Yau threefold with canonical singularities, and (as we spell out in Proposition 3.1) when all edges of Δ have lattice length one the singular points of X are precisely cones $C(F)$ over the two-dimensional faces F of Δ.

Putting these together: *if a reflexive 4-polytope Δ has a two-dimensional face F with unit edges whose associated cone $C(F)$ is non-smoothable, then the generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ is a Calabi–Yau threefold that admits no smoothing* (Corollary 4.3; by Remark **??** only the edges of F itself need lattice length one). The point is that non-smoothable faces are easy to recognize (a finite combinatorial test, Theorem 2.2) and easy to plant (a small search over completions of an embedded polygon, §5).

Our two main examples are the following. Write $e_1, \dots, e_4$ for the standard basis of $\mathbb{Z}^4$.

Theorem 1.1 (= Theorem 5.1)**.** *Let*

$$\Delta_A = \mathrm{conv}\{(0,0,1,0),\, (-2,-1,1,0),\, (-2,-2,1,0),\, (-1,-2,1,0),\, e_4,\, (1,1,-1,-1)\} \subset \mathbb{Z}^4.$$

Then Δ_A is reflexive, and the generic anticanonical hypersurface $X_A \subset \mathbb{P}_{\Delta_A}$ is a Calabi–Yau threefold whose singular locus consists of exactly three points, each analytically isomorphic to the anticanonical cone over the Hirzebruch surface $\mathbb{F}_1$. Every deformation of X_A over a reduced irreducible base preserves these three singular germs; in particular X_A admits no smoothing. The maximal projective crepant partial resolution $\widehat{X}_A$ is a smooth Calabi–Yau threefold with $(h^{1,1}, h^{2,1}) = (5, 101)$.

The local model here is precisely Gross's exceptional germ: the cone over $\mathbb{F}_1$ is *rigid in the reduced sense*. Its miniversal base is a fat point, with $\dim T^1 = 1$ but no positive-dimensional deformations [13, Example 2.19, §5.5]. Since it is the cone over a non-simplicial polygon, it is not a quotient singularity, so X_A is not covered by the classical $\frac{1}{3}(1,1,1)$ examples. Compact Calabi–Yau threefolds containing such points are implicit in the primitive-contraction literature (see [13, 15, 14], where the smoothable exceptional types are systematically smoothed and the $\mathbb{F}_1$ case is excluded); we are not aware of an explicit compact example in print, and Δ_A provides one lying in the Kreuzer–Skarke classification [16].

The second example exhibits what we believe is a new phenomenon on compact Calabi–Yau threefolds.

Theorem 1.2 (= Theorem 5.3)**.** *Let*

$$\Delta_B = \mathrm{conv}\{(0,0,1,0),\ (-2,-1,1,0),\ (-3,-2,1,0),\ (-2,-3,1,0),\ (-1,-2,1,0),\ e_4,\ (1,1,-1,-1)\} \subset \mathbb{Z}^4.$$

Then Δ_B is reflexive, and the generic anticanonical hypersurface $X_B \subset \mathbb{P}_{\Delta_B}$ is a Calabi–Yau threefold whose singular locus consists of exactly three points, each analytically isomorphic to the cone $C(Q_B)$ over the pentagon Q_B that is the Minkowski sum of the $\frac{1}{3}(1,1,1)$-triangle and a unit segment. Each singular germ has one-dimensional reduced miniversal base, so each singular point of X_B admits a non-trivial one-parameter local deformation; nevertheless the miniversal base of each germ has no smoothing component, and X_B admits no smoothing. The maximal projective crepant partial resolution $\widehat{X}_B$ is a smooth Calabi–Yau threefold with $(h^{1,1}, h^{2,1}) = (9, 81)$.

Thus X_B is a canonical Calabi–Yau threefold all of whose singularities admit genuine deformations — there is no infinitesimal or local obstruction visible at first order, and the germs move — and yet no deformation ever smooths it. The unique local deformation direction pulls the pentagon apart into its Minkowski summands, trading each pentagon germ for a $\frac{1}{3}(1,1,1)$ germ, which is then rigid: the singularity "cascades" to its rigid core and stops. All previously known non-smoothable canonical Calabi–Yau examples that we are aware of are locally rigid (quotient points as in [13], or the $\mathbb{F}_1$-cone above).

Both examples have exactly three singular points, and this is forced by the shape of the construction: each is built by adjoining two further vertices to Q, placed as a two-dimensional face at height one, and the requirement that 0 be interior then separates the two facets through that face far enough to give $\ell(F^\circ) \geq 2$ (Lemma **??**). Adjoining a *third* vertex removes the constraint and allows $\ell(F^\circ) = 1$, giving a Calabi–Yau threefold whose singular locus is a *single* non-smoothable point (Theorem 5.5; the same device produces a Calabi–Yau threefold with a single $\frac{1}{3}(1,1,1)$ point, where the classical X_9 has three). The deformable analogue exists as well: a systematic scan of the Kreuzer–Skarke classification [16] locates a 10-vertex reflexive polytope whose hypersurface has a single singular point of the pentagon type $C(Q_B)$ (Theorem 5.7). This hypersurface is a compact Calabi–Yau threefold with one singular point which deforms non-trivially and yet is never absorbed by a smoothing.

The two theorems are instances of a general construction: realize a prescribed polygon as a two-dimensional face of a reflexive 4-polytope. We state this as Proposition 3.1 and Corollary 4.3; it turns the smoothing question for a large class of compact Calabi–Yau threefolds into polygon combinatorics. As bookkeeping for that combinatorics we record in §2 a complete census of the isolated Gorenstein toric threefold singularities whose polygon has an edge representation with coordinates in $[-2, 2]$: there are 217 classes up to equivalence, comprising the ordinary double point, 8 rigid classes, 79 deformable-but-non-smoothable classes, and 129 smoothable classes. The five classes whose polygon has one interior lattice point are exactly the anticanonical cones over the five smooth toric del Pezzo surfaces, and the census reproduces the classical deformation-theoretic data of these cones [13, §5.5] exactly (Remark 2.6); this both validates the computation and places the new examples in context: non-smoothability is common among Gorenstein toric threefold singularities (87 of the 216 canonical classes in the census; here and below, "canonical" counts exclude the census's unique terminal class, the ordinary double point), while the del Pezzo cone catalogue sees only its smallest stratum.

Our approach is the Calabi–Yau analogue, one dimension up, of the program of Petracci and Corti–Hacking–Petracci on (non-)smoothability of Gorenstein toric *Fano* threefolds [19, 5], where the local models are the same cones $C(F)$, attached to the facets of a reflexive 3-polytope. In our setting the local-to-global step is simpler than in loc. cit.: because the relevant local models are non-smoothable outright, no almost-flatness or bundle-cohomology obstruction is needed. The versal base of the germ already has no smoothing component, and this kills all global smoothings (Lemma 4.1).

All computations in this paper (the census, the face inventories of the example polytopes, the dual-edge point counts, the completion-search statistics and the Kreuzer–Skarke scan of §6, and the Hodge numbers) are implemented in six short exact-arithmetic computer programs, validated against the classical cases recalled in the text (the quintic; $X_9 \subset \mathbb{P}(1,1,1,3,3)$; the del Pezzo cone catalogue; Altmann's hexagon; and, for the scan, the Hodge numbers recorded in the Kreuzer–Skarke data itself). They are included with the arXiv submission as ancillary files and are public, together with all data and results, at `https://github.com/bwuebben/calabi-yau-smoothability`.

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2. Local models: cones over polygons

2.1. **The dictionary.** Let $N \cong \mathbb{Z}^3$ and let $Q \subset N_{\mathbb{R}}$ be a lattice polygon placed at height one, i.e. $Q \subset \{\langle u^*, \cdot\rangle = 1\}$ for a primitive $u^* \in M = N^\vee$. Write σ_Q for the cone over Q and

$$C(Q) \;=\; \operatorname{Spec} \mathbb{C}[\sigma_Q^\vee \cap M].$$

Every Gorenstein toric affine threefold without torus factors arises this way, and $C(Q)$ depends only on Q up to affine unimodular equivalence of the plane $\{\langle u^*, \cdot\rangle = 1\}$, i.e. up to $\mathbb{Z}^2 \rtimes GL_2(\mathbb{Z})$ acting on Q. We freely identify Q with a polygon in $\mathbb{Z}^2$. The basic dictionary (see e.g. [4]) is:

- $C(Q)$ has an *isolated* singularity if and only if every edge of Q has lattice length one (*unit edges*); an edge of lattice length $m \geq 2$ produces a curve of transverse A_{m-1}-singularities.
- $C(Q)$ is canonical and Gorenstein; it is *terminal* if and only if Q has no lattice points other than its vertices, and it is smooth if and only if Q is a standard triangle (unimodular image of $\operatorname{conv}\{(0,0),(1,0),(0,1)\}$).
- For a unit-edge k-gon, Pick's theorem gives $i = A - k/2 + 1$ interior lattice points, where A is the euclidean area.

Since translations do not change the edges, a unit-edge polygon is equivalent data to its multiset $E(Q) = \{e_1, \ldots, e_k\}$ of primitive edge vectors, listed counterclockwise, with $\sum e_i = 0$; and $GL_2(\mathbb{Z})$-equivalence of polygons is $GL_2(\mathbb{Z})$-equivalence of edge multisets.

2.2. **Deformations: the trichotomy.** Minkowski decompositions of unit-edge polygons are transparent at the level of edge multisets.

Lemma 2.1. *Let Q be a unit-edge polygon with edge multiset $E(Q)$. Minkowski decompositions $Q = R_1 + \cdots + R_r$ into lattice polytopes (up to translation of the summands) correspond bijectively to partitions $E(Q) = E_1 \sqcup \cdots \sqcup E_r$ into non-empty zero-sum sub-multisets. Under this correspondence a summand is a unit segment if and only if its part is a pair $\{v, -v\}$, and a standard triangle if and only if its part is a triple $\{a, b, c\}$ with $a + b + c = 0$ and $|\det(a, b)| = 1$.*

Proof. If $Q = R + S$, then each edge of Q decomposes as the sum of the faces of R and S with the same outer normal, and lattice lengths add; since the edges of Q are primitive, each edge comes entirely from R or from S. Hence $E(Q) = E(R) \sqcup E(S)$, and inductively any decomposition partitions $E(Q)$ into zero-sum parts. Conversely, a zero-sum sub-multiset of primitive vectors, sorted counterclockwise, closes up to a convex lattice polygon (a segment when the part is $\{v, -v\}$), and the Minkowski sum of the polygons associated to the parts of a partition has edge multiset $E(Q)$; since a convex polygon is determined by its edge multiset up to translation, that sum is Q. The last statement is immediate: a triangle with primitive edges $a, b, c = -a - b$ is unimodular if and only if $|\det(a, b)| = 1$. □

The deformation theory of $C(Q)$ is completely described by two results.

Theorem 2.2 (Altmann [2]; Corti–Filip–Petracci [4, §5.1])**.** *Let Q be a unit-edge lattice polygon, not a standard triangle, and let $V = C(Q)$.*

(1) *The irreducible components of the reduced miniversal base space of V are in bijection with the maximal Minkowski decompositions of Q into lattice summands; each component is smooth, and the component associated to a decomposition with r summands has dimension $r - 1$.*
(2) *The smoothing components are exactly those associated to decompositions all of whose summands are unit segments and standard triangles.*

Consequently, with the notation of Lemma 2.1 (this repackaging, and the labels below, are ours), exactly one of the following holds:

(R) (rigid) $E(Q)$ *has no proper non-empty zero-sum sub-multiset. Then the reduced miniversal base is a point: V has no positive-dimensional deformations, and is not smoothable.*
(D) (deformable, non-smoothable) $E(Q)$ *has a proper zero-sum sub-multiset, but admits no partition into pairs $\{v, -v\}$ and unimodular triples. Then V deforms non-trivially but no deformation has smooth general fibre.*
(S) (smoothable) $E(Q)$ *admits such a partition; the number of such partitions equals the number of smoothing components of the miniversal base.*

For the dimensions in (1) and the concentration of T^1 in the Gorenstein degree see also [1]; for unit-edge k-gons one has $\dim T^1 = k - 3$. (In the count of smoothing components, unit segments and standard triangles are Minkowski-indecomposable, so decompositions into them are automatically *maximal* and index components in Altmann's parametrization.) Filip [7] re-derives and extends part (2) to non-isolated Gorenstein toric singularities via 0-mutable Laurent polynomials.

Example 2.3. The two smallest illustrations of the three classes:

- The $\frac{1}{3}(1, 1, 1)$ singularity $= \mathbb{C}^3/\mu_3$ is the cone over the triangle T with $E(T) = \{(-2, -1), (1, -1), (1, 2)\}$; no proper subset sums to zero, so it is rigid, recovering Schlessinger [20].
- The ordinary double point is the cone over the unit square, $E = \{\pm(1, 0), \pm(0, 1)\}$: one partition into two segment pairs, hence smoothable with a single smoothing component ($xy - zw = t$).
- The pentagon $Q_B = T + s$, the Minkowski sum of T with a unit segment s, has $E(Q_B) = \{(-2, -1), (-1, -1), (1, -1), (1, 1), (1, 2)\}$. Its only proper zero-sum sub-multisets are the pair $\{(-1, -1), (1, 1)\}$ and the complementary triple $E(T)$, which is not unimodular ($\det = 3$). So Q_B has exactly one non-trivial maximal decomposition,

$T + s$: the reduced miniversal base of $C(Q_B)$ is a smooth curve, and there is no smoothing component; $C(Q_B)$ is deformable but not smoothable. (Here $\dim T^1 = 2$, so the miniversal base is in addition non-reduced.)

The next lemma describes the general fibre of the homogeneous deformation attached to a Minkowski decomposition; we use it in Remark 5.4 to identify what the local deformations of our second example do to the germ.

Lemma 2.4 (General fibre of the Altmann family). *Let $Q = R_0 + \cdots + R_r$ be a Minkowski decomposition of a unit-edge polygon into lattice summands, let $f_0, \dots, f_r$ be the standard basis of $\mathbb{Z}^{r+1}$, and let $\widetilde{V}$ be the Gorenstein toric $(3+r)$-fold associated to the* Cayley cone $\widetilde{\sigma} = \operatorname{Cone}\big(\bigcup_i R_i \times \{f_i\}\big) \subset \mathbb{R}^2 \oplus \mathbb{R}^{r+1}$*; write $u_0, \dots, u_r$ for the dual basis of $f_0, \dots, f_r$, viewed as characters of $\widetilde{V}$, and let*

$$\pi = \big(\chi^{u_0} - \chi^{u_1},\ \dots,\ \chi^{u_0} - \chi^{u_r}\big) \colon \widetilde{V} \longrightarrow \mathbb{C}^r, \qquad \pi^{-1}(0) \cong C(Q),$$

be Altmann's homogeneous deformation attached to the decomposition [1] (*cf.* [4, Remark 6.2]). *Then:*

1. *the general fibre of π has exactly one singular point of analytic type $C(R_i)$ for each two-dimensional non-smooth summand R_i, and no other singular points;*
2. *for the pentagon $Q_B = T + s$ of Example 2.3, the family π dominates the positive-dimensional component of* $\operatorname{Def} C(Q_B)$*, whose general fibre therefore has a single singular point, of type $\frac{1}{3}(1,1,1)$.*

Proof. (1) For two-dimensional R_i the orbit closure of the face $\operatorname{Cone}(R_i \times \{f_i\})$ is an r-dimensional affine toric variety whose character lattice is freely generated by the restrictions of the u_j, $j \neq i$, while χ^{u_i} vanishes on it; so for generic $t \in \mathbb{C}^r$ the r equations of the fibre $\pi^{-1}(t)$ determine on it the single reduced point $\chi^{u_0} = t_i$, $\chi^{u_j} = t_i - t_j$ (with the convention $t_0 = 0$, which makes the formula uniform in i), lying in the open orbit, and the implicit-function argument in the proof of Proposition 3.1(2) below identifies the germ of $\pi^{-1}(t)$ there with $(C(R_i), 0)$, on the chart cut out by the characters χ^{u_j}, $j \neq i$, which form a basis of the saturated annihilator of $\operatorname{Cone}(R_i \times \{f_i\})$ and exhibit it as $C(R_i) \times (\mathbb{C}^*)^r$. On the orbit closure of a face of $\widetilde{\sigma}$ meeting two distinct summand heights, some equation of $\pi^{-1}(t)$ restricts to $0 = t_i$ or $t_i = t_j$, so the general fibre avoids these strata. The remaining strata are the open orbit and the orbits of the proper faces τ of the cones $\operatorname{Cone}(R_i \times \{f_i\})$ (together with, for a *segment* summand R_i, the two-dimensional cone $\operatorname{Cone}(R_i \times \{f_i\})$ itself, which the same computation below covers), and there the fibre is smooth. Over the open orbit this holds because $u_0, \dots, u_r$ extend to a basis of the character lattice. Along the orbit of a proper face τ the ambient $\widetilde{V}$ is itself smooth: τ is spanned either by a single ray (v, f_i) with v a vertex of R_i, which is primitive, or by the rays over an edge $[v, v']$ of R_i, and since the edge is unit the pair (v, f_i), $(v' - v, 0)$ extends to a lattice basis. The functionals u_j, $j \neq i$ (including u_0 when $i \neq 0$), vanish on τ and form part of a basis of the saturated lattice $\tau^\perp \cap M$, so on the smooth chart $U_\tau \cong \mathbb{C}^{\dim \tau} \times (\mathbb{C}^*)^{3+r-\dim \tau}$ the characters χ^{u_0} and χ^{u_j}, $j \neq i$, may be taken as coordinates of the torus factor, while χ^{u_i} is divisible by a nonconstant monomial in the $\mathbb{C}^{\dim \tau}$-coordinates and hence vanishes along the orbit together with its derivatives in the torus coordinates. The Jacobian of the r equations of $\pi^{-1}(t)$ with respect to these torus coordinates is therefore triangular with diagonal entries ± 1, and the fibre is a smooth complete intersection near the orbit.

(2) By Example 2.3 the reduced miniversal base of $C(Q_B)$ is a smooth curve and $T + s$ is the unique non-trivial maximal decomposition, with $r = 1$. By (1) the general fibre of

π has a single singular point of type $C(T) = \frac{1}{3}(1,1,1)$, not $C(Q_B)$; the family is therefore non-trivial, and its classifying map to the miniversal base is non-constant and dominates the curve. Hence the general fibre over the positive-dimensional component of $\operatorname{Def} C(Q_B)$ is the general fibre of π: a single $\frac{1}{3}(1,1,1)$ point. The only available deformation of the germ thus trades the pentagon cone for the rigid quotient cone, after which no further deformation is possible. □

2.3. **A census, and the del Pezzo catalogue.** Both tests in Theorem 2.2 are finite, so the trichotomy can be tabulated. We enumerated all unit-edge polygons whose (counterclockwise) edge vectors have coordinates in $[-2,2]$ (all k-gons with $3 \le k \le 16$ occur) and reduced modulo $GL_2(\mathbb{Z})$.

Proposition 2.5. *Up to equivalence there are exactly* 217 *unit-edge polygons whose* (*counterclockwise*) *primitive edge vectors can be chosen with all coordinates in* $[-2,2]$*, excluding the standard triangle. They comprise: the unit square (the ordinary double point, the unique terminal class, smoothable);* 8 *rigid classes;* 79 *deformable-but-non-smoothable classes; and* 129 *smoothable classes. The* 8 *rigid classes are, listed as* $[k, i;\, E(Q)]$*:*

$$\begin{gathered}
[3,1;\ (-2,-1),(1,-1),(1,2)]\\
[4,1;\ (-2,-1),(0,-1),(1,0),(1,2)]\\
[4,2;\ (-2,-1),(2,-1),(1,2),(-1,0)]\\
[4,2;\ (-2,-1),(2,-1),(1,1),(-1,1)]\\
[5,2;\ (-2,-1),(2,-1),(1,1),(0,1),(-1,0)]\\
[5,3;\ (-2,-1),(-1,-1),(2,-1),(1,2),(0,1)]\\
[5,4;\ (-2,-1),(0,-1),(2,-1),(1,2),(-1,1)]\\
[6,5;\ (-2,-1),(-1,-1),(2,-1),(1,0),(1,2),(-1,1)]
\end{gathered}$$

Proof. By computer; the paragraph below records exactly what the computation establishes and how it is checked. □

The results depend on the census in the following way. The type of a class and its number of smoothing components are $GL_2(\mathbb{Z})$-invariants of the edge multiset (Lemma 2.1), hence independent of how the reduction to equivalence classes is performed; they are checked against the classical cases (the conifold, the rigid $\frac{1}{3}(1,1,1)$ cone, Altmann's dP_6 hexagon with its two smoothing components, and the del Pezzo catalogue of Remark 2.6). The class count 217 depends on the equivalence test, and here a finite test suffices for a proof: if $g \in GL_2(\mathbb{Z})$ carries one census edge multiset onto another, pick two independent edges u, v of the first, with images u', v'; then $g = [\,u' \,|\, v'\,]\operatorname{adj}[\,u \,|\, v\,]/\det[\,u \,|\, v\,]$ with all eight entries of the two outer matrices in $[-2,2]$ and $|\det[\,u \,|\, v\,]| \ge 1$, so every entry of g is bounded by 8. The pairwise reduction, re-run over all $GL_2(\mathbb{Z})$-elements with entries bounded by 8, merges no two of the 217 representatives; the count is therefore exact. Note that rigidity is not confined to polygons with few edges: for every k there are rigid unit-edge k-gons, e.g. with $E(Q) = \{(1,a_1),\ldots,(1,a_{k-1}),\,(-(k-1),-\sum a_j)\}$ for distinct a_j with $\gcd(k-1,\sum a_j) = 1$: any zero-sum subset must contain the last vector and then all the others.

Remark 2.6 (del Pezzo cross-check). The census contains exactly five classes with $i = 1$ interior point. If Q is unit-edge with $i = 1$ interior lattice point, then Q is reflexive (after centering) and one checks that $\sigma_Q^\vee$ is the cone at height one over the polar dual Q°; hence

$$C(Q) \;\cong\; \operatorname{Spec} \textstyle\bigoplus_{m\ge 0} H^0\big(S,\, \omega_S^{-m}\big)\,,$$

the anticanonical cone over the toric surface S with moment polygon Q°. For the five census classes the surfaces S are smooth: they are precisely the five smooth toric del Pezzo surfaces, and the census reproduces the classical deformation theory of their cones [13, §5.5] (which rests on [2]):

S	$\deg S$	$[k,i]$	$\dim T^1 = k-3$	type
$\mathbb{P}^2$	9	$[3,1]$	0	rigid
$\mathbb{F}_1$	8	$[4,1]$	1	rigid (fat point)
$\mathbb{P}^1\times\mathbb{P}^1$	8	$[4,1]$	1	smoothable, 1 component
dP_7	7	$[5,1]$	2	smoothable, 1 component
dP_6	6	$[6,1]$	3	smoothable, 2 components

(The pair $[k,i]$ alone does not determine the class: $\mathbb{F}_1$ and $\mathbb{P}^1\times\mathbb{P}^1$ share $[4,1]$ and differ in type.) In particular the second rigid class of Proposition 2.5, Q_A with $E(Q_A) = \{(-2,-1), (0,-1),(1,0),(1,2)\}$, satisfies $Q_A^\circ =$ the moment polygon of $(\mathbb{F}_1, -K)$, so $C(Q_A)$ *is the anticanonical cone over* $\mathbb{F}_1$, the exceptional germ of [13, Theorem 5.8]. The deformable-but-non-smoothable classes all have $i \geq 2$; they are cones over polarized toric surfaces that are not anticanonically polarized del Pezzo surfaces, and so lie outside the catalogue above.

3. Two-dimensional faces of reflexive polytopes and Batyrev hypersurfaces

Let $\widetilde{N} \cong \mathbb{Z}^4$ and let $\Delta \subset \widetilde{N}_\mathbb{R}$ be a reflexive polytope: a lattice polytope with 0 in its interior whose polar dual $\Delta^\circ = \{u : \langle u, v\rangle \geq -1\ \forall v \in \Delta\}$ is again a lattice polytope. Let $\mathbb{P}_\Delta$ be the toric fourfold of the face fan of Δ; it is Gorenstein Fano, $-K_{\mathbb{P}_\Delta}$ is ample and globally generated, and its space of sections has the monomial basis $\{\chi^m : m \in \Delta^\circ \cap \widetilde{M}\}$ [3, 6], where $\widetilde{M} = \widetilde{N}^\vee$ is the dual lattice. Throughout, Δ is the *fan* polytope (Batyrev's Δ^*), so our Δ° is his Δ, the Newton polytope of the anticanonical sections.

For a face $G \preceq \Delta$ write σ_G for the cone over G, $V(\sigma_G) \subset \mathbb{P}_\Delta$ for the corresponding orbit closure, and $G^\circ \preceq \Delta^\circ$ for the dual face. If F is a two-dimensional face, then σ_F is a three-dimensional cone, $V(\sigma_F)$ is a toric curve $\cong \mathbb{P}^1$, and F° is an edge of Δ°; we write $\ell(F^\circ)$ for its lattice length.

Normalization convention. When listing example data we describe each facet of Δ by the primitive integral functional u normalized so that the facet lies on $\{\langle u, \cdot\rangle = 1\}$ (we say u is *at level one*); the corresponding vertex of Δ° is then $-u$, and the dual face G° of a face G is spanned by the *negatives* of the facet functionals through G. Since lattice length is invariant under $u \mapsto -u$, we compute $\ell(F^\circ)$ directly on the listed functionals.[1] The vertices $-u$ of Δ° dual to the two facets of Δ containing F restrict to the sublattice $N_F = \widetilde{N} \cap \mathbb{R}\sigma_F$ as a single integral height function equal to -1 on F; its negative (the common restriction of the two listed functionals) is a height function equal to $+1$ on F, so (σ_F, N_F) is the Gorenstein cone over the lattice polygon F in the height-one convention of §2, and we write $C(F)$ for the associated toric threefold, computed in the lattice induced on the affine span of F.

Proposition 3.1. *Let $\Delta \subset \mathbb{Z}^4$ be reflexive, all of whose edges have lattice length one, and let $X \in |-K_{\mathbb{P}_\Delta}|$ be generic. Then:*

1. *X is a Calabi–Yau threefold with Gorenstein canonical singularities.*
2. *For each two-dimensional face $F \preceq \Delta$ with $C(F)$ singular, X meets the curve $V(\sigma_F)$ in exactly $\ell(F^\circ)$ points, all in the open orbit, and at each such point p the germ (X, p) is analytically isomorphic to $(C(F), 0)$.*

[1] This matches the normalization used by the ancillary programs.

(3) *X is smooth away from these points.*

Proof. (1) is Batyrev's theorem [3, §4].

(2) Fix such an F. The restriction map on anticanonical sections sends χ^m to a non-zero section on $V(\sigma_F)$ exactly when $m \in F^\circ \cap \widetilde{M}$: indeed χ^m vanishes identically on $V(\sigma_F)$ unless $\langle m, \cdot\rangle$ attains its minimum -1 on F. The restricted line bundle is the toric line bundle on $V(\sigma_F) \cong \mathbb{P}^1$ with one-dimensional moment polytope F°, of degree $\ell(F^\circ)$ [6, §6.3], and the lattice points of F° restrict to the full monomial basis of $H^0(\mathbb{P}^1, \mathcal{O}(\ell(F^\circ)))$. Hence for generic X the intersection $X \cap V(\sigma_F)$ consists of $\ell(F^\circ)$ distinct reduced points avoiding the two torus-fixed points.

For the local structure, split $\widetilde{N} \cong N_F \oplus \mathbb{Z}$; then the open toric chart of σ_F is $U_{\sigma_F} \cong C(F) \times \mathbb{C}^*$, with $V(\sigma_F) \cap U_{\sigma_F} = \{0\} \times \mathbb{C}^*$ [6, Prop. 3.3.9]. Trivialize $-K$ on this chart and embed $C(F) \subset \mathbb{C}^r$ by a generating set of $\sigma_F^\vee \cap M_F$ (writing $M_F = N_F^\vee$); the defining equation of X extends to a function $\widetilde{f}(z,t)$ on $\mathbb{C}^r \times \mathbb{C}^*$. At a point $p = (0, t_0)$ of the transverse intersection above, $\widetilde{f}(0,\cdot)$ has a simple zero at t_0, so $\partial\widetilde{f}/\partial t(p) \neq 0$, and by the implicit function theorem $\{\widetilde{f} = 0\}$ is, near p, the graph of an analytic function $t = \varphi(z)$. Intersecting with $C(F) \times \mathbb{C}^*$ and projecting along the graph yields an analytic isomorphism $(X,p) \cong (C(F), 0)$.

(3) $\operatorname{Sing}\mathbb{P}_\Delta$ is the union of the orbit closures of the non-smooth cones of the face fan. Cones over vertices are smooth: every vertex of a reflexive polytope is primitive, since a lattice point of the dual facet of Δ° would otherwise pair with it outside $\mathbb{Z}$. The full-dimensional cones' smooth loci are irrelevant for a generic hypersurface: since $-K_{\mathbb{P}_\Delta}$ is globally generated, Bertini gives that X is smooth away from $\operatorname{Sing}\mathbb{P}_\Delta$, and X avoids the finitely many torus-fixed points (the section χ^{u_Γ}, for u_Γ the vertex of Δ° dual to a facet Γ, is non-vanishing at the fixed point of σ_Γ). For a two-dimensional cone σ_e over an edge e of Δ with vertices v, w: since $\ell(e) = 1$ and v lies at height 1 with respect to the (restriction of the) supporting functional of any facet containing e, the pair $(v, w - v)$ is a basis of $\widetilde{N} \cap \mathbb{R}\sigma_e$, so σ_e is smooth. Hence, under the all-unit-edges hypothesis, $\operatorname{Sing}\mathbb{P}_\Delta$ meets X only in the curves $V(\sigma_F)$ of part (2), and there only at the listed points. ☐

Remark 3.2. If Δ has an edge e of lattice length $m \geq 2$, the same analysis shows that X acquires a curve of transverse A_{m-1}-singularities along $V(\sigma_e)$; we impose the all-unit-edges hypothesis only to keep the singular locus zero-dimensional. The local statement (2) at a *unit-edge* face F is unaffected by longer edges elsewhere: the points of X on $V(\sigma_F)$ remain isolated singular points with germ $C(F)$, and part (1) holds for any reflexive Δ, so Corollary 4.3 below applies to every reflexive Δ possessing a unit-edge face of type (R) or (D), whether or not the remaining edges are unit.

4. Local-to-global: no smoothing

Lemma 4.1. *Let X be a reduced projective variety with an isolated singular point p such that no irreducible component of the reduced miniversal base space of the germ (X,p) has smooth general fibre. Then for every flat proper family $\mathcal{X} \to (T,0)$ over a reduced irreducible pointed base with $\mathcal{X}_0 \cong X$, every fibre $\mathcal{X}_t$ for t near 0 is singular. In particular X admits no smoothing.*

If moreover the reduced miniversal base of (X,p) is a point, then the induced family of germs at p is trivial: every $\mathcal{X}_t$ has a singular point with germ isomorphic to (X,p).

Proof. By Grauert [11], the isolated singularity (X,p) has a miniversal deformation $\mathcal{S} \to (S,0)$, which we represent on a suitable Milnor representative; versality holds for analytic families over reduced base germs after shrinking [12, II.1]. The relative singular locus $\Sigma \subset \mathcal{S}$ is closed analytic, and (after shrinking) the projection $\Sigma \to S$ is finite: its central fibre is the isolated point p, so on a Milnor representative, whose boundary the nearby singular loci avoid, the projection is proper with finite fibres. Its image is therefore a closed analytic subset $D \subseteq S$, the locus of singular fibres. A component of S is a smoothing component precisely when it is not contained in D; by hypothesis there are none, so $D = S$ and *every* fibre of the miniversal family is singular.

Now let $\mathcal{X} \to (T, 0)$ be as in the statement and restrict it to a Milnor representative around p; after shrinking T this restriction is induced from $\mathcal{S} \to S$ by a base-change map $h : (T, 0) \to (S, 0)$. Its fibres are fibres of the miniversal family, hence singular; so $\mathcal{X}_t$ has a singular point near p for every t near 0.

For the final statement: if S_{red} is a point, then h factors through S_{red} because T is reduced, i.e. h is constant, and the induced family of germs is the pullback of the central fibre. □

Remark 4.2. Lemma 4.1 is deliberately the crudest possible local-to-global statement: it requires no Calabi–Yau hypothesis and no obstruction theory, because the local model already forbids smooth nearby fibres. For the much finer relationship between local and global deformations of a Calabi–Yau threefold with isolated canonical singularities, in particular when local smoothings do exist and the question is whether they are realized by global deformations, see Gross [13, §§1–2] and Namikawa's stratified local moduli [17].

Corollary 4.3. *Let Δ be a reflexive 4-polytope with unit edges possessing a two-dimensional face F such that the polygon F is of type* (R) *or* (D) *in Theorem 2.2. Then the generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ is a Calabi–Yau threefold with canonical Gorenstein singularities which admits no smoothing.*

Proof. By Proposition 3.1, X has $\ell(F^\circ) \geq 1$ points with germ $C(F)$; by Theorem 2.2 the miniversal base of this germ has no smoothing component; by Lemma 4.1 no deformation of X over a reduced irreducible base (in particular no smoothing over a disc) has smooth general fibre. □

5. The examples

Both examples plant a non-smoothable polygon Q from §2 as a two-dimensional face of a reflexive 4-polytope of the simple shape $\operatorname{conv}(Q \times \{(1,0)\} \cup \{e_4,\ w\})$, found by a small computer search over the completing vertex w. (Placing Q in the plane $x_3 = 1$, $x_4 = 0$ embeds it in a saturated affine sublattice, so the face polygon carries the intrinsic lattice structure of Q; translations of Q within that plane and the normalization of the first completing vertex to e_4 are unimodular coordinate changes.)

5.1. A Calabi–Yau threefold with three $\mathbb{F}_1$-cone points.

Theorem 5.1. *Let Q_A be the rigid quadrilateral with $E(Q_A) = \{(-2,-1), (0,-1), (1,0), (1,2)\}$, embedded with vertices $(0,0), (-2,-1), (-2,-2), (-1,-2)$, and let*

$$\Delta_A = \operatorname{conv}\big(Q_A \times \{(1,0)\} \ \cup \ \{e_4,\ (1,1,-1,-1)\}\big) \subset \mathbb{Z}^4.$$

Then:

(1) *Δ_A is reflexive with 6 vertices and 6 facets, and all 13 edges of Δ_A have lattice length one. Its 13 two-dimensional faces consist of the face $F_A = Q_A \times \{(1,0)\}$ and 12 standard triangles.*

(2) *The generic anticanonical hypersurface* $X_A \subset \mathbb{P}_{\Delta_A}$ *is a Calabi–Yau threefold whose singular locus consists of exactly* $\ell(F_A^\circ) = 3$ *points, at each of which the germ of* X_A *is the anticanonical cone over* $\mathbb{F}_1$*.*

(3) *Every deformation of* X_A *over a reduced irreducible base preserves the three singular germs up to analytic isomorphism. In particular* X_A *admits no smoothing.*

(4) *Batyrev's formulas give* $(h^{1,1}, h^{2,1}) = (5, 101)$*, hence Euler number* -192*, for the maximal projective crepant partial resolution* $\widehat{X}_A$*.*

Proof. (1) is a finite verification. The six facet inequalities $\langle u, \cdot \rangle \le 1$ of Δ_A have primitive normals

$$(0,0,1,1),\ (0,0,1,-2),\ (-3,6,1,1),\ (6,-3,1,1),\ (-3,0,-5,1),\ (0,-3,-5,1),$$

each at level exactly 1, which is reflexivity; the face F_A is the intersection of Δ_A with the two supporting hyperplanes of the first two normals. The remaining two-dimensional faces admit a uniform description: writing $w = (1,1,-1,-1)$ for the second completing vertex, the segment $[e_4, w]$ is an edge of Δ_A, and the twelve triangles are the joins of the four edges of F_A with e_4, the joins of those edges with w, and the joins of the four vertices of F_A with the edge $[e_4, w]$; each is unimodular, and every edge of Δ_A has lattice length one (all verified in exact arithmetic). The dual edge F_A° is spanned by the negatives of the first two normals (the convention of §3), of lattice length $\gcd\big((0,0,1,1) - (0,0,1,-2)\big) = 3$.

(2) By Proposition 3.1, the singular points of X_A are: for F_A, three points with germ $C(Q_A)$; for the twelve standard-triangle faces, none, as their cones are smooth. By Remark 2.6, $C(Q_A)$ is the anticanonical cone over $\mathbb{F}_1$.

(3) Q_A is rigid: $E(Q_A)$ has no proper zero-sum sub-multiset (a direct check of the 14 proper non-empty subsets). By Theorem 2.2(R) the reduced miniversal base of each germ is a point, and Lemma 4.1 applies, including its final statement.

(4) With $|\Delta_A \cap \mathbb{Z}^4| = 8$ and $|\Delta_A^\circ \cap \mathbb{Z}^4| = 130$ lattice points and the face data above, Batyrev's formulas [3] for the Hodge numbers of $\widehat{X}_A$ give $h^{1,1} = 5$ and $h^{2,1} = 101$.[2] □

Remark 5.2. The three germs of X_A have $\dim T^1 = 1$ but fat-point miniversal bases: X_A has first-order deformations at each singular point, all obstructed. This is exactly the local model excluded in Gross's smoothability theorem for primitive type II contractions [13, Theorem 5.8, §5.5], here realized three times on an explicit compact Calabi–Yau threefold in the Kreuzer–Skarke list.

5.2. **A deformable but non-smoothable Calabi–Yau threefold.**

Theorem 5.3. *Let* $Q_B = T + s$ *be the type-*(D) *pentagon of Example 2.3, embedded with vertices* $(0,0), (-2,-1), (-3,-2), (-2,-3), (-1,-2)$*, and let*

$$\Delta_B = \operatorname{conv}\big(Q_B \times \{(1,0)\} \ \cup\ \{e_4,\ (1,1,-1,-1)\}\big) \subset \mathbb{Z}^4.$$

Then:

(1) Δ_B *is reflexive with* 7 *vertices and* 7 *facets; all* 16 *edges have lattice length one; its* 16 *two-dimensional faces are* $F_B = Q_B \times \{(1,0)\}$ *and* 15 *standard triangles.*

(2) *The generic anticanonical* $X_B \subset \mathbb{P}_{\Delta_B}$ *is a Calabi–Yau threefold whose singular locus consists of exactly* $\ell(F_B^\circ) = 3$ *points, each with germ* $C(Q_B)$*.*

[2] As validation, the same computation gives $(1, 101)$ for the quintic simplex and $(4, 112)$ for $X_9 \subset \mathbb{P}(1,1,1,3,3)$.

(3) *Each singular germ of X_B admits a non-trivial one-parameter deformation: its reduced miniversal base is a smooth curve. Nevertheless the miniversal base has no smoothing component, and X_B admits no smoothing.*
(4) $(h^{1,1}, h^{2,1}) = (9, 81)$ *for the maximal projective crepant partial resolution, hence Euler number* -144.

Proof. (1) The seven facet normals are

$$(0,0,1,1),\ (0,0,1,-2),\ (-3,6,1,1),\ (6,-3,1,1),$$
$$(-3,3,-2,1),\ (-1,-1,-4,1),\ (3,-3,-2,1),$$

each at level 1; the rest is finite verification as in Theorem 5.1(1), with F_B° again of length 3; the fifteen triangles are again the joins of the five edges of F_B with each completing vertex and of the five vertices of F_B with the edge joining the completing vertices. (2) is Proposition 3.1 plus the face inventory. (3) is Example 2.3 combined with Theorem 2.2 and Lemma 4.1. (4) is computed as in Theorem 5.1(4), with $|\Delta_B \cap \mathbb{Z}^4| = 10$, $|\Delta_B^\circ \cap \mathbb{Z}^4| = 104$. □

Remark 5.4. By Lemma **??**, the available local deformation at each singular point of X_B replaces the pentagon cone by a $\frac{1}{3}(1,1,1)$ point. Thus, to the extent that these local deformations are realized by global ones, X_B moves in a family whose neighbouring members are Calabi–Yau threefolds with three rigid quotient points, which then admit no further deformation of their germs. In every case the singularities never disappear. We are not aware of a previous compact Calabi–Yau example in which all singular points are locally deformable and yet no smoothing exists; the known non-smoothable examples (quotient points, and the $\mathbb{F}_1$-cone of Theorem 5.1) are locally rigid.

5.3. **A single non-smoothable point.** The examples above have three singular points because the dual edge of the prescribed face has lattice length 3. Within the two-vertex family this cannot be pushed below 2.

Lemma 5.5 (two adjoined vertices force $\ell(F^\circ) \geq 2$)**.** *Let Q be a unit-edge lattice polygon, let $\Delta = \operatorname{conv}\big(Q \times \{(1,0)\} \cup \{e_4, w\}\big) \subset \mathbb{Z}^4$ be reflexive, and let $F = Q \times \{(1,0)\}$. Then F is a two-dimensional face of Δ and $\ell(F^\circ) \geq 2$.*

Proof. F is a face: reflexivity puts 0 in the interior, which (as shown below) forces $w_3 \leq -1$, so $u_0 = (0,0,1,0)$ takes the value 1 on $Q \times \{(1,0)\}$ and less on e_4 and w, i.e. it supports Δ exactly along F. Every facet of Δ containing F has normal of the form $u_t = (0,0,1,t)$ with $t \in \mathbb{Z}$ (these are exactly the integral functionals restricting to 1 on the affine span of F), and the two facets through F are the extreme supporting members of this pencil: for $t_{\min} < t < t_{\max}$ the face of u_t is $\operatorname{face}(u_{t_{\min}}) \cap \operatorname{face}(u_{t_{\max}}) = F$, so the interval $[t_{\min}, t_{\max}]$ is exactly F° and $\ell(F^\circ) = t_{\max} - t_{\min}$. With completing vertices e_4 and $w = (w_1, w_2, w_3, w_4)$ the bound $\ell(F^\circ) \geq 2$ is forced, as follows. All vertices of $Q \times \{(1,0)\}$ have $x_3 = 1$ and $x_4 = 0$, and e_4 has $x_3 = 0$, $x_4 = 1$; for the origin to be interior some vertex must have negative third coordinate and some vertex negative fourth coordinate, so $w_3 \leq -1$ and $w_4 \leq -1$. The member u_t supports Δ if and only if $\langle u_t, e_4\rangle = t \leq 1$ and $\langle u_t, w\rangle = w_3 + t w_4 \leq 1$; since $w_4 < 0$ the second constraint bounds t from below only, so $t_{\max} = 1$, attained on the facet through e_4. The facet at the lower extreme must contain a vertex outside the plane of F, necessarily w, so $t_{\min} = (1 - w_3)/w_4$, an integer with positive numerator (≥ 2) and negative denominator; hence $t_{\min} \leq -1$ and $\ell(F^\circ) = 1 - t_{\min} \geq 2$. □

A third adjoined vertex in $\{x_3 = 1\}$ removes the interiority constraint that forced $w_3 \leq -1$, and realizes $t_{\min} = 0$:

Theorem 5.6. *Let Q_A be embedded as in Theorem 5.1 and let*

$$\Delta_C = \operatorname{conv}\big(Q_A \times \{(1,0)\} \ \cup\ \{e_4,\ (1,1,-1,0),\ (-2,-2,1,-1)\}\big) \subset \mathbb{Z}^4.$$

Then:

(1) *Δ_C is reflexive with 7 vertices and 10 facets; all 18 edges have lattice length one; its 21 two-dimensional faces are $F_C = Q_A \times \{(1,0)\}$ and 20 standard triangles; and the dual edge F_C°, spanned by the negatives of the facet functionals $(0,0,1,0)$ and $(0,0,1,1)$, has lattice length 1.*

(2) *The generic anticanonical hypersurface $X_C \subset \mathbb{P}_{\Delta_C}$ is a Calabi–Yau threefold whose singular locus is a* single *point, at which the germ is the anticanonical cone over $\mathbb{F}_1$. Every deformation of X_C over a reduced irreducible base preserves this germ; in particular X_C admits no smoothing.*

(3) *The maximal projective crepant partial resolution $\widehat{X}_C$ has $(h^{1,1}, h^{2,1}) = (4, 108)$, hence Euler number -208.*

Proof. The ten facet normals are

$$\begin{array}{lllll} (0,0,1,1), & (0,0,1,0), & (-2,0,-3,0), & (-2,0,-3,1), & (-2,4,1,-4), \\ (-2,4,1,1), & (0,-2,-3,0), & (0,-2,-3,1), & (4,-2,1,-4), & (4,-2,1,1), \end{array}$$

each at level exactly 1; the face inventory, edge lengths, and the dual edge F_C° are finite verifications as in Theorem 5.1(1). Parts (2) and (3) follow exactly as in Theorem 5.1, with $|\Delta_C \cap \mathbb{Z}^4| = 9$, $|\Delta_C^\circ \cap \mathbb{Z}^4| = 141$ in Batyrev's formulas. □

Remark 5.7. The same pair of completing vertices applied to the $\frac{1}{3}(1,1,1)$ triangle (embedded with vertices $(0,0), (-2,-1), (-1,-2)$) gives a 6-vertex reflexive polytope whose generic anticanonical hypersurface is a Calabi–Yau threefold with a *single* $\frac{1}{3}(1,1,1)$ point, with resolution Hodge numbers $(3,111)$ and $\chi = -216$; the classical $X_9 \subset \mathbb{P}(1,1,1,3,3)$ has three such points.

The rigid examples above leave one configuration outstanding: a single singular point of the *deformable* type. Our completion searches realize $\ell(F^\circ) = 1$ for type-(D) faces only on polytopes carrying additional singular faces; the example below was instead found in the Kreuzer–Skarke classification itself, by the scan described in §6.

Theorem 5.8. *Let*

$$\begin{aligned} \Delta_D = \operatorname{conv}\{&(1,0,0,0),\ (0,1,0,0),\ (0,-1,0,0),\ (0,0,1,0),\ (-3,1,-1,0), \\ &(0,1,1,0),\ (0,0,0,1),\ (-3,2,1,-1),\ (-2,2,2,-1),\ (1,0,1,1)\} \subset \mathbb{Z}^4. \end{aligned}$$

Then:

(1) *Δ_D is reflexive with 10 vertices and 15 facets; all 30 edges have lattice length one; its 35 two-dimensional faces are 34 standard triangles and one pentagon F_D, which is $GL_2(\mathbb{Z})$-equivalent to the type-(D) pentagon $Q_B = T + s$ of Theorem 5.3; and the dual edge F_D°, spanned by the negatives of the facet functionals $(0,1,0,1)$ and $(-1,-1,1,1)$, has lattice length 1.*

(2) *The generic anticanonical hypersurface $X_D \subset \mathbb{P}_{\Delta_D}$ is a Calabi–Yau threefold whose singular locus is a* single *point, with germ $C(Q_B)$: the reduced miniversal base of the singularity is a smooth curve, so the singular point of X_D admits a non-trivial one-parameter local deformation, and yet X_D admits no smoothing.*

(3) *The maximal projective crepant partial resolution $\widehat{X}_D$ has $(h^{1,1}, h^{2,1}) = (8, 118)$, hence Euler number -220.*

Proof. All statements in (1) are finite verifications in exact arithmetic as in Theorem 5.1(1): the fifteen facet normals are

$$\begin{array}{lllll} (0,1,0,1), & (-1,-1,1,1), & (1,1,0,0), & (1,1,0,-1), & (1,0,1,-1), \\ (1,1,-1,1), & (1,1,-1,-3), & (1,1,-3,1), & (1,1,-3,-5), & (1,-1,1,-1), \\ (1,-1,1,-3), & (1,-1,-1,1), & (1,-1,-1,-7), & (1,-1,-5,1), & (1,-1,-5,-11), \end{array}$$

each at level exactly 1; the pentagon face F_D has vertices $5,7,8,9,10$ in the order listed, its edge multiset in the induced lattice is $\{(1,0),(3,2),(-1,0),(-3,-1),(0,-1)\}$, and the unimodular map $\left(\begin{smallmatrix} -1 & 2 \\ -1 & 1 \end{smallmatrix}\right)$ carries it to $E(Q_B)$; the first two facet normals listed are those of the two facets through F_D, and the dual edge they determine has lattice length 1. Part (2) follows from Proposition 3.1, Example 2.3, Theorem 2.2 and Lemma 4.1 exactly as in Theorem 5.3; part (3) is Batyrev's formulas with $|\Delta_D \cap \mathbb{Z}^4| = 15$, $|\Delta_D^\circ \cap \mathbb{Z}^4| = 152$. □

Remark 5.9. Theorems 5.5 and 5.7 together answer the minimality question for both non-smoothable types: one rigid point, or one deformable point, on a compact Calabi–Yau three-fold, each realized inside the Kreuzer–Skarke classification. The scan that produced Δ_D found three 10-vertex polytopes whose unique *non-smoothable* face is of type (D) with dual edge of length one; Δ_D is the one whose other faces are all smooth (the other two carry additional ordinary smoothable singular points). Over the full classification there are exactly four polytopes whose generic hypersurface has a *single* singular point of type (D), with 10, 11, 11 and 12 vertices (Δ_D being the minimal one), and in all four the germ is $C(Q_B)$; see §6.

6. Further remarks and questions

6.1. **How common is this?** The completion search that produced Δ_A and Δ_B is the simplest possible: complete the embedded polygon by two vertices, one of which can be normalized to e_4. This family realizes 18 of the 87 non-smoothable census classes of Proposition 2.5 as two-dimensional faces of reflexive 4-polytopes (both examples above use the same completing vertex $(1,1,-1,-1)$), and the count is stable: enlarging the coordinate box for the completing vertex from $[-3,3]$ to $[-5,5]$ produces no new classes, and every class realized this way has at most 6 interior lattice points. Completing by *three* vertices instead (as in Theorem 5.5, with both free vertices ranging over the box $[-2,2]^4$) raises the count to 32 of the 87: all 23 classes with $i \leq 6$ embed, together with 9 classes with $7 \leq i \leq 10$.

A complementary and ultimately decisive approach is to scan the Kreuzer–Skarke classification itself. We have carried out the test of Theorem 2.2 on *all* 473,800,776 reflexive 4-polytopes [16] (on the completeness of the enumeration see the end of this subsection).

Proposition 6.1 (the scan)**.** *Of the* 473,800,776 *reflexive* 4*-polytopes, exactly* 39,175,536 *(that is,* 8.27%*) carry a two-dimensional face with primitive edges whose cone is not smoothable* (*on the completeness of the enumeration behind these counts, see the end of this subsection*)*. For each of these the generic anticanonical hypersurface is a Calabi–Yau threefold admitting no smoothing, by Corollary* 4.3 (*via Remark* **??** *for those carrying longer edges elsewhere*)*.*

Proof. By Theorem 2.2 the type of $C(F)$ depends only on the edge multiset of F, so the test is a finite computation on each polytope; it was run on the whole classification, and the surviving counts are recorded below. □

In more detail we find:

- 8.27% (that is, 39,175,536 of them) carry a unit-edge non-smoothable two-dimensional face, so their generic anticanonical hypersurfaces are non-smoothable Calabi–Yau threefolds by Corollary 4.3 (via Remark **??** for the polytopes carrying longer edges elsewhere); the fraction rises from 11.7% at 5 vertices to a peak of 14.0% at 9 and then declines steadily with the vertex count, to below 1% beyond 22 vertices;
- faces of type (D) appear from 7 vertices on (as they must: a unit-edge triangle is indecomposable and a decomposable unit-edge quadrilateral splits into two segment pairs and is smoothable, so a type-(D) polygon has at least five edges, and a pentagonal two-face together with the two further vertices needed to span dimension four already requires seven vertices), 497,434 face occurrences in all;
- combining with the completion searches above, exactly 65 of the 87 census classes are realized as two-dimensional faces of reflexive 4-polytopes. The 22 classes that never occur are all of type (D) and all have $i \geq 11$ interior points, so the occurrence of a class decays with its size and, beyond a threshold, ceases entirely, an obstruction we do not explain;
- unit-edge non-smoothable faces *outside* the range of Proposition 2.5 also occur (rigid polygons with an edge coordinate exceeding 2; those we inspected have at most five edges), so the classes of Proposition 2.5 do not exhaust the phenomenon.

This answers the first natural quantitative question (what fraction of the Batyrev landscape is non-smoothable) completely, and the second (which local models occur) for the classes of Proposition 2.5; the unit-edge non-smoothable faces outside that range occur but remain to be classified. The single-point examples are similarly constrained: among all reflexive 4-polytopes with unit edges whose generic hypersurface has a *single* singular point, and that point of type (D), the point is the pentagon cone $C(Q_B)$ of Theorem 5.7 in *every* case (the four such polytopes on the list all give $C(Q_B)$); no other type-(D) germ occurs as the sole singularity of a Batyrev threefold. (Type-(D) germs of other classes do occur as the unique *non-smoothable* point of a hypersurface that also carries smoothable singular points; it is only as a lone singularity that $C(Q_B)$ is forced.)

Completeness of the enumeration. We scanned 473,800,775 polytopes through the per-vertex-count copy of the Kreuzer–Skarke database that we used, whose files cover 5 to 33 vertices, and the one remaining polytope separately. Granting that the copy's entries are distinct members of the classification (the polytope counts and Hodge data recorded for each vertex number were validated throughout the scan), the one polytope it omits is the classification's unique member with more than 33 vertices, and we identify it: it is the 36-vertex product of two reflexive hexagons, verified directly.[3] That polytope's 48 two-dimensional faces are 36 unit parallelograms and 12 hexagons of dP_6 type, all of type (S) with dual edge of length one, so it contributes to none of the counts above.

Question 6.2. The faces with non-unit edges (whose germs are non-isolated, governed by Filip's 0-mutability criterion [7] rather than by Theorem 2.2) deserve their own count.

6.2. **The smoothable faces and the local-to-global problem.** Corollary 4.3 uses only the non-smoothable local models. In the opposite regime, where all two-dimensional faces are of type (S), local smoothings exist at every singular point, and the question becomes whether they can be realized simultaneously by a global deformation; for ordinary double points this is Friedman's criterion [8], and for Calabi–Yau threefolds with canonical singularities the local-to-global obstruction theory of [13, 17] applies. A combinatorial criterion on Δ for

[3]Ancillary file `missing_polytope.py`.

global smoothability of the Batyrev hypersurface, in the spirit of the Fano-threefold results of [19, 5], is the natural continuation of this note; it is the subject of the companion paper [**?**], which smooths the points over a type-(S) face whenever its dual edge has lattice length at least two, shows the threshold sharp, and compares the resulting criterion with the census of Batyrev–Kreuzer [**?**] of the all-conifold regime.

6.3. **Mirror symmetry.** Corti–Filip–Petracci conjecture that the smoothing components of $\operatorname{Def} C(Q)$ correspond to the 0-mutable Laurent polynomials with Newton polygon Q [4], a mirror-theoretic statement. What the presence of a type (R) or (D) face of Δ, and hence the non-smoothability of X, means for the mirror family of the Batyrev pair (Δ, Δ°) is taken up in [**?**], which classifies the pairs whose *both* hypersurfaces have isolated singularities and finds that a type-(D) face never occurs among them.

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